

CO2 - Assessment Tool 2

CODE: DSA0506-QUERRY PROCESSSSING FOR DATASCIENCE

NAME: V.TEJASWINI

REG NO:192424360

-

Question 5: University Course Registration Database (10 Marks) A university maintains the following database schema: • Student (Student_ID, Student_Name, Department) • Course (Course_ID, Course_Name, Credits) • Enrollment (Enrollment_ID, Student_ID, Course_ID, Semester) Analyze the schema and write a Python program to retrieve the student name, course name, and semester for all students enrolled in courses carrying more than 3 credits.

Code:

```
import sqlite3

# Connect to SQLite database

conn = sqlite3.connect("university.db")

cursor = conn.cursor()

# Create Student table

cursor.execute("""

CREATE TABLE IF NOT EXISTS Student(

    Student_ID INTEGER PRIMARY KEY,

    Student_Name TEXT,

    Department TEXT

)

""")

# Create Course table

cursor.execute("""

CREATE TABLE IF NOT EXISTS Course(

    Course_ID INTEGER PRIMARY KEY,

    Course_Name TEXT,
```

```

        Credits INTEGER
    )
    """
# Create Enrollment table
cursor.execute("""
CREATE TABLE IF NOT EXISTS Enrollment(
    Enrollment_ID INTEGER PRIMARY KEY,
    Student_ID INTEGER,
    Course_ID INTEGER,
    Semester TEXT,
    FOREIGN KEY(Student_ID) REFERENCES Student(Student_ID),
    FOREIGN KEY(Course_ID) REFERENCES Course(Course_ID)
)
""")
# Insert sample data
cursor.execute("INSERT INTO Student VALUES (1,'Rahul','CSE')")
cursor.execute("INSERT INTO Student VALUES (2,'Priya','ECE')")
cursor.execute("INSERT INTO Student VALUES (3,'Arun','IT')")
cursor.execute("INSERT INTO Course VALUES (101,'Python Programming',4)")
cursor.execute("INSERT INTO Course VALUES (102,'Database Systems',3)")
cursor.execute("INSERT INTO Course VALUES (103,'Machine Learning',5)")
cursor.execute("INSERT INTO Enrollment VALUES (1,1,101,'Semester 1')")
cursor.execute("INSERT INTO Enrollment VALUES (2,2,102,'Semester 2')")
cursor.execute("INSERT INTO Enrollment VALUES (3,3,103,'Semester 1')")
conn.commit()
# Query
print("Students enrolled in courses with more than 3 credits:\n")

```

```

cursor.execute("""
SELECT Student.Student_Name,
       Course.Course_Name,
       Enrollment.Semester
FROM Student
JOIN Enrollment
ON Student.Student_ID = Enrollment.Student_ID
JOIN Course
ON Course.Course_ID = Enrollment.Course_ID
WHERE Course.Credits > 3
""")
rows = cursor.fetchall()
for row in rows:
    print("Student Name :", row[0])
    print("Course Name :", row[1])
    print("Semester   :", row[2])
    print()
conn.close()

```

output:

```

... Customers with account balance greater than ₹1,00,000:

Customer Name : Ramesh
Account No    : 1001
Balance       : ₹ 150000.0

Customer Name : Kiran
Account No    : 1003
Balance       : ₹ 200000.0

```

2. Question 6: Banking Database (10 Marks) A bank maintains the following database schema: • Customer (Customer_ID, Customer_Name, City) • Account (Account_No, Customer_ID, Account_Type, Balance) • Transaction (Transaction_ID, Account_No, Transaction_Date, Amount) Analyze the schema and write a Python program to retrieve the customer name, account number, and balance for customers whose account balance exceeds ₹1,00,000

Code:

```
import sqlite3

# Connect to SQLite database
conn = sqlite3.connect("bank.db")

cursor = conn.cursor()

# Create Customer table
cursor.execute("""
CREATE TABLE IF NOT EXISTS Customer(
    Customer_ID INTEGER PRIMARY KEY,
    Customer_Name TEXT,
    City TEXT
)
""")

# Create Account table
cursor.execute("""
CREATE TABLE IF NOT EXISTS Account(
    Account_No INTEGER PRIMARY KEY,
    Customer_ID INTEGER,
```

```
Account_Type TEXT,  
Balance REAL,  
FOREIGN KEY(Customer_ID) REFERENCES Customer(Customer_ID)  
)  
""")
```

Create Transaction table

```
cursor.execute("""  
CREATE TABLE IF NOT EXISTS Transaction_Details(  
Transaction_ID INTEGER PRIMARY KEY,  
Account_No INTEGER,  
Transaction_Date TEXT,  
Amount REAL,  
FOREIGN KEY(Account_No) REFERENCES Account(Account_No)  
)  
""")
```

Insert sample data

```
cursor.execute("INSERT INTO Customer VALUES (1,'Ramesh','Chennai')")  
cursor.execute("INSERT INTO Customer VALUES (2,'Anitha','Hyderabad')")  
cursor.execute("INSERT INTO Customer VALUES (3,'Kiran','Bangalore')")  
cursor.execute("INSERT INTO Account VALUES (1001,1,'Savings',150000)")  
cursor.execute("INSERT INTO Account VALUES (1002,2,'Current',75000)")  
cursor.execute("INSERT INTO Account VALUES (1003,3,'Savings',200000)")  
cursor.execute("INSERT INTO Transaction_Details VALUES (1,1001,'2026-08-01',5000)")  
cursor.execute("INSERT INTO Transaction_Details VALUES (2,1002,'2026-08-02',3000)")
```

```
cursor.execute("INSERT INTO Transaction_Details VALUES (3,1003,'2026-08-03',7000)")
```

```
conn.commit()
```

```
# Query
```

```
print("Customers with account balance greater than ₹1,00,000:\n")
```

```
cursor.execute("""
```

```
SELECT Customer.Customer_Name,
```

```
       Account.Account_No,
```

```
       Account.Balance
```

```
FROM Customer
```

```
JOIN Account
```

```
ON Customer.Customer_ID = Account.Customer_ID
```

```
WHERE Account.Balance > 100000
```

```
""")
```

```
rows = cursor.fetchall()
```

```
for row in rows:
```

```
    print("Customer Name :", row[0])
```

```
    print("Account No   :", row[1])
```

```
    print("Balance      : ₹", row[2])
```

```
    print()
```

```
conn.close()
```

output:

```
convert2000()
```

```
... Students enrolled in courses with more than 3 credits:
```

Output

```
Student Name : Rahul  
Course Name  : Python Programming  
Semester     : Semester 1
```

```
Student Name : Arun  
Course Name  : Machine Learning  
Semester     : Semester 1
```